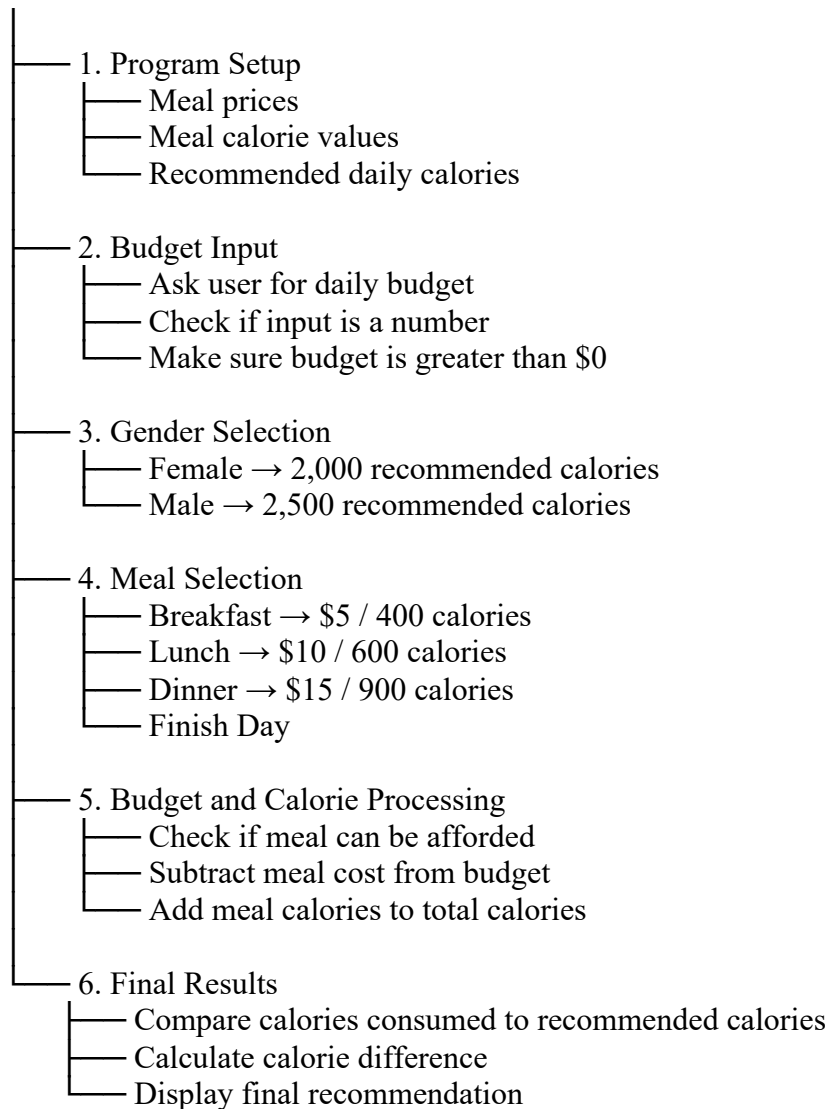


# Budget Calorie Meal Planner

## Modular Program Hierarchy

### Program Hierarchy

#### Budget Calorie Meal Planner



## **Module Descriptions**

### **1. Program Setup**

This section stores the constant values used throughout the program. These include the price and calorie amount for breakfast, lunch, and dinner, along with the recommended daily calories for women and men.

### **2. Budget Input**

This part asks the user to enter a daily meal budget. A loop and try/catch statement are used to make sure the user enters a number greater than zero. Invalid values cause an error message to appear.

### **3. Gender Selection**

The program asks the user to select female or male. The selection determines whether the program uses a recommended daily calorie amount of 2,000 or 2,500 calories.

### **4. Meal Selection**

The user selects breakfast, lunch, dinner, or Finish Day. A switch statement assigns the correct cost and calorie value to the selected meal.

### **5. Budget and Calorie Processing**

Before accepting a meal, the program checks whether the user has enough money remaining. If the meal can be purchased, the cost is removed from the budget and the meal's calories are added to the total calories consumed.

### **6. Final Results**

When meal planning ends, the program compares the user's total calories with the recommended calorie amount. It calculates the difference and provides a final result to the user.

## **Module Interaction**

The program follows this basic flow:

Program Setup → Budget Input → Gender Selection → Meal Selection → Budget/Calorie Processing → Final Results

The Meal Selection and Budget/Calorie Processing sections repeat until the user finishes the day or no longer has enough money to continue.